En mall (och kort introduktion) till LaTeX

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April 10, 2026

Abstract

Detta dokument ger dig smakprov på vad man kan göra med LaTeX. Genom att titta på och ändra i koden som ligger till grund för detta dokument kan du förhoppningsvis snabbt komma igång med att skriva snygga rapporter.

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1 Introduktion

When modelling bear populations, age determination constitutes a key component for understanding and predicting population dynamics. More recently, DNA-based methods have been developed (Nakamura et al., 2025), which may provide age estimates with better error characteristics for population level inference.

Both the magnitude and the distribution of age determination errors may differ between methods as well as depend on the true age of the individual. For example, very young individuals cannot have their age underestimated below zero, whereas older individuals may exhibit larger errors. Additionally, accurately estimating the age of individuals near the threshold between non-reproductive and reproductive age is particularly important for population-level analyses.

It is thus important to assess how different forms of age determination error affect estimated population age distributions, and to what extent conclusions drawn from observed age data may be sensitive to these errors.

To investigate how age determination error affects estimated population age distributions, a simulation-based approach will be used. First, a baseline age distribution representing the true population age structure will be constructed using a simple population matrix model, inspired by models presented in Caswell (2001) and parameterized with data from Støen m.fl. (2021).

Next, age determination error will be modelled as a random variable whose distribution may depend on the true age of the individual. To this purpose we will use n different distribution. A baseline case where there error is

Several plausible error distributions will be considered, with their parameters chosen so that the standard deviation of the distribution matches roughly the RMSE of the observed residuals for each age determination method.

As a final step, a deconvolution matrix will be constructed to examine whether it is possible to recover the true age distribution from observed age data, given knowledge of the age determination error distribution. This matrix can then be used to assess the accuracy of population-level inference for different age determination methods.

2 Theory

2.1 Matrix population modells

To obtain a suitable idea of the population age distribution over time we use a population matrix model Caswell 2001 which we will describe shortly in this section. The age class $i = 1, 2, ...m$ is the discretization of the age x to age-class i with $i - 1 \leq x \leq i$. The number of individuals for each age class i is then given by $n_1, n_2, ..., n_m$.

Following Caswell 2001, Eq. 2.1 the number of n_1 individuals at time $t + 1$ can be projected with

$$n_1(t + 1) = f_1 n_1(t) + f_2 n_2(t) + ... + f_i n_i(t)$$

Here f_i denotes the reproductive rate per-capita. That is the average number of offspring an individual of age class i has. For $i > 1$ the number of individuals in an age class n_i at time $t + 1$ is simply given by their survival probaility $(s_{i-1})(n_{i-1})$ at the previous timestep t. Together we can write the equation in matrix form for all n_i at time $t + 1$.

$$\begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ \vdots \\ n_m \end{pmatrix} (t+1) = \begin{pmatrix} f_1 & f_2 & f_3 & \cdots & f_m \\ s_1 & 0 & 0 & \cdots & 0 \\ 0 & s_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & s_{m-1} & 0 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ \vdots \\ n_m \end{pmatrix} (t)$$

2.2 Computing age distribution from a Leslie matrix

The stable age distribution of a population can be obtained from the Leslie matrix $\mathbf{A}$ by computing its dominant eigenvalue λ_1 and the corresponding right eigenvector $\mathbf{w}$. This eigenvector gives the relative proportions of individuals in each age class at equilibrium, and can be normalized so that the entries sum to 1.

The existence of a real, non-negative eigenvalue λ_1 that is greater than or equal in magnitude to all other eigenvalues λ_i of a non-negative, primitive matrix $\mathbf{A}$ is guaranteed by the Perron-Frobenius theorem Caswell 2001.

If $\mathbf{A}$ is primitive, then as $t \to \infty$, the population vector $\mathbf{n}_t$ satisfies

$$\frac{\mathbf{n}_t}{\sum_i n_{i,t}} \to \mathbf{w},$$

where $\mathbf{w}$ is the normalized right eigenvector corresponding to λ_1. This represents the **stable age distribution** of the population.

2.3 Age determination errors

Once the stable age distribution of the population has been obtained, we simulate the effect of age determination errors.

Let ϵ denote the measurement error in age, which may depend on the true age X:

$$\epsilon \mid X \sim N(0, f(X)^2).$$

The observed age Y is then given by

$$Y = X + \epsilon.$$

Applying this error to the theoretical distribution produces the observed age distribution. This can be represented using a misclassification matrix M, where M_{ij} denotes the probability that an individual of true age class i is observed in age class j.

If $\mathbf{p}_x$ is the vector of true age proportions and $\mathbf{p}_y$ the observed proportions, then

$$\mathbf{p}_y = M\mathbf{p}_x.$$

2.4 misclassification matrices

For discrete age classes, define a misclassification matrix M such that

$$M_{ij} = P(Y = j \mid X = i)$$

is the probability that an individual of true age class i is observed in age class j. Let $\mathbf{p}_x$ be the vector of true age proportions (from the stable distribution $\mathbf{w}$), and $\mathbf{p}_y$ the observed age proportions. Then the observed distribution is

$$\mathbf{p}_y = M\mathbf{p}_x.$$

2.5 Stable age distribution and age measurement errors

Let $\mathbf{A}$ be a Leslie matrix describing the age-structured population dynamics. The population vector at time t, $\mathbf{n}_t$, evolves as

$$\mathbf{n}_{t+1} = \mathbf{A}\mathbf{n}_t.$$

Stable age distribution. The dominant eigenvalue λ_1 of $\mathbf{A}$ and its corresponding right eigenvector $\mathbf{w}$ satisfy

$$\mathbf{A}\mathbf{w} = \lambda_1 \mathbf{w}, \quad \mathbf{w} \geq 0.$$

By the Perron–Frobenius theorem Caswell 2001, λ_1 is real and non-negative, and as $t \to \infty$, the normalized population vector converges to the **stable age distribution**:

$$\frac{\mathbf{n}_t}{\sum_i n_{i,t}} \longrightarrow \mathbf{w}, \quad \text{with } \sum_i w_i = 1.$$

Age measurement errors. Let X denote the true age of an individual, and suppose age determination is subject to measurement error ϵ:

$$Y = X + \epsilon, \quad \epsilon \mid X \sim N(0, f(X)^2),$$

where $f(X)$ specifies the standard deviation of the error as a function of true age.

Summary. This formulation allows us to combine the theoretical population structure (from the Leslie matrix) with realistic measurement error in age, producing an observed age distribution that accounts for both **population dynamics** and **age-determination uncertainty**.

2.6 Matrix inversion & condition number

[TODO: Does this need a source reference?]

2.7 Thikanov regularization

TODO: Expand this section.
Include SVD formulation, derivation of the filter factor $\frac{\sigma_i}{\sigma_i^2+\lambda}$, and discussion of how regularization trades bias for stability in ill-posed inverse problems.

2.8 Expectation-Maximization algorithm (EM-algorithm)

The EM algorithm is a general-purpose method for computing maximum likelihood estimates in the presence of incomplete or latent data (McLachlan and Krishnan 2008). Let $\mathbf{Y}$ denote a discrete random vector corresponding to the observed (incomplete) data, and $\mathbf{X}$ a discrete random vector representing the complete data. If $\mathbf{X}$ were fully observable, we could directly compute the complete-data log-likelihood function $\ell_c(\theta)$ based on a realization x.

However, in practice we only observe $y \in \mathcal{Y}$, the sample space of $\mathbf{Y}$. The complete-data sample space $\mathcal{X}$ is related to $\mathcal{Y}$ through a many-to-one mapping, i.e., a surjection from $\mathcal{X}$ to $\mathcal{Y}$, where multiple elements $x \in \mathcal{X}$ correspond to the same observed value y.

The probability mass function of $\mathbf{Y}$ is therefore given by

$$p_Y(y;\theta) = \sum_{x \in \mathcal{X}(y)} p_X(x;\theta), \tag{1}$$

where $\mathcal{X}(y) = \{x \in \mathcal{X} : h(x) = y\}$.

Since the complete data $\mathbf{X}$ is not observable, the complete-data log-likelihood $\ell_c(\theta)$ cannot be evaluated directly. Instead, the EM algorithm considers its conditional expectation given the observed data y and a current parameter estimate $\theta^{(t)}$:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}\left[\ell_c(\theta) \mid Y = y\right] \tag{2}$$

which corresponds to the Expectation step (E-step) of the EM algorithm. In the Maximization step (M-step), the parameter is updated by maximizing this function:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}). \tag{3}$$

The process is then iterated upon until $l(\theta^{t+1}) - l(\theta^t) <= k$ for some arbitrary k value.

Monotonicity . A key property for the algorithm to work is for the likelihood to be monotonically increasing for each iteration. $L(\theta^{t+1}) \geq L(\theta^t))$ which is discussed in Section 3.2 of McLachlan and Krishnan 2008.

2.9 Application of the EM algorithm to the deconvolution problem

Let $X = (Z, Y)$ denote our complete data, that is in our deconvolution problem the observed age Y and the corresponding true age Z. The PMF for X is then:

$$p_X(x = (z_i, y_j); \theta_Z) = p_i M_{ij} \tag{4}$$

where M is the missclasification matrix defined in 2.4 and for a vector of n independent observations $\mathbf{x} = (z_1, y_1), (z_2, y_2), ..., (z_n, y_n)$ we obtain

$$p_X(\mathbf{x}; \theta_Z) = \prod_i \prod_j (p_i M_{ij})^{n_{ij}} \tag{5}$$

where $n_i j = \sum_{k=1} 1\{X_k = i, Y_k = j\}$ is a random variable with the number of counts that were observed as j given they were i.

The log-likelihood becomes:

$$\ell_c(\theta_Z) = \sum_i \sum_j n_{ij} \log(p_i M_{ij}) = \sum_i \sum_j n_{ij} \log M_{ij} + \sum_i \sum_j n_{ij} \log p_i \quad (6)$$

In accordance with equation (2) we take the conditional expectation on $n_i j$ with an initial parameter θ^0 conditioned on Y

$$\mathbb{E}_{\theta^{(0)}}\left[\ell_c(\theta) \mid Y = j\right] = \mathbf{E}_{\theta^{(0)}}\left[\sum_i \sum_j n_{ij} \log(p_i M_{ij}) \mid Y = j\right] = \sum_i \sum_j \mathbf{E}_{\theta^{(0)}}\left[n_{ij} \mid Y = j\right] \log(p_i M_{ij}) \quad (7)$$

The conditional expectation $E[n_i j | y = j]$ is the expected number of bears of age i missclasified as age j given the total amount of bears observed as age j.

Taking expectation:

$$\mathbb{E}[n_{ij} \mid Y = y] = \sum_{k=1}^{n} \mathbb{E}\left[\mathbf{1}\{Z_k = i, Y_k = j\} \mid Y = y\right] \quad (8)$$

Now conditioning on the observed Y_k:

$$= \sum_{k:Y_k=j} P\left(Z_k = i \mid Y_k = j, \theta^{(0)}\right) \quad (9)$$

Let $n_j = \#\{k : Y_k = j\}$ denote the number of observations with $Y = j$. Then:

$$\mathbb{E}[n_{ij} \mid Y = y] = n_j \cdot P\left(Z = i \mid Y = j, \theta^{(0)}\right) \quad (10)$$

Using Bayes' theorem:

$$P\left(Z = i \mid Y = j, \theta^{(0)}\right) = \frac{p_i^{(0)} M_{ij}}{\sum_{i'} p_{i'}^{(0)} M_{i'j}} \quad (11)$$

Thus:

$$\mathbb{E}[n_{ij} \mid Y = y] = n_j \cdot \frac{p_i^{(0)} M_{ij}}{\sum_{i'} p_{i'}^{(0)} M_{i'j}} \quad (12)$$

Maximization-step

2.10 Evaluation method

Total Varation (TV) To quantify the difference between two discrete probability distributions Z and Q over the same finite sample space $\mathcal{X}$, we use the *Total Variation (TV) distance*:

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P(x) - Q(x)|.$$

The TV distance ranges from 0 (distributions are identical) to 1 (distributions have no overlap). Intuitively, it represents the maximum fraction of probability mass that must be reallocated to transform P into Q. In this thesis, we use TV distance to assess how convolution and sampling affect the observed age distributions, as well as to evaluate the accuracy of deconvolution methods.

[TODO: Does this need a source reference?]

KL-divergence **[TODO: unfinished section, needs reference.]**

3 Methods / Simulation Setup

3.1 Simulated Age Distributions

A Leslie matrix L of dimension 20×20 was used to model the age-structured population. The first row represents the expected number of offspring produced by individuals in each age class, calculated as the product of three components for age class i:

$$F_i = s_i \cdot p_{\mathrm{coy},i} \cdot \mathrm{litter}_i, \tag{13}$$

where:

- s_i is the survival probability from age i to $i + 1$,
- $p_{\mathrm{coy},i}$ is the probability of conceiving at age i,
- litter_i is the expected litter size at age i.

The survival and reproduction parameters were estimated from published graphs Norwegian Institute for Nature Research (NINA) n.d. The full data vectors used to construct the Leslie matrix are provided in Appendix 6.1.

3.2 Error terms

To simulate plausible errors in age determination, we considered four different error models. Let Z denote the true age, and e the error term added to Z in the convolution. We model the error as conditionally normal:

$$e \mid Z \sim \mathcal{N}\big(\mu = f(Z), \sigma^2\big),$$

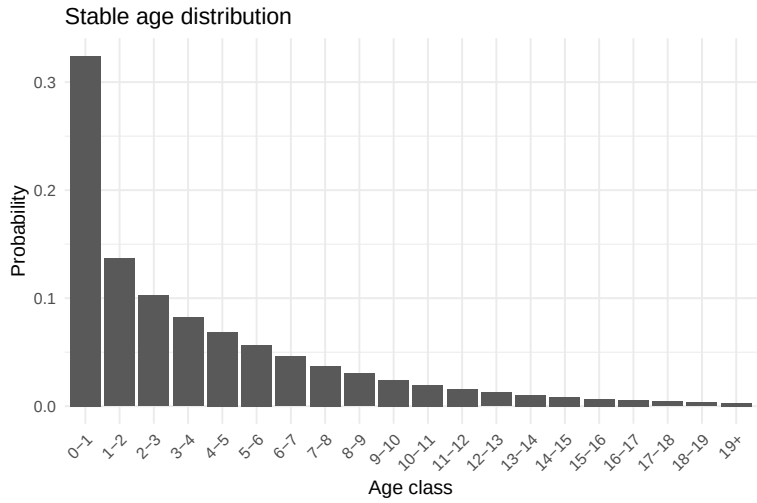


Figure 1: Stable age distribution obtained from the normalized dominant eigenvector of the Leslie matrix (see Section 2.2 for derivation).

and discretize the resulting values to the integer age bins (1–20) to match the observed age categories. Negative values were excluded, and the remaining probabilities were rescaled by $1/(1 - F(e < 0))$. Values greater than 20 were assigned to the 19+ age class.

The following four specifications for $f(Z)$ (mean) and σ (standard deviation) were considered:

Model A: Model A Unbiased, low variance: $f(Z) = 0, \quad \sigma = 1$

Model B: Model B Unbiased, high variance: $f(Z) = 0, \quad \sigma = 4$

Model C: Model C Age-dependent bias, low variance: $f(Z) = \dfrac{4}{Z + 0.5}$, $\sigma = 1$

Model D: Model D Age-dependent bias, high variance: $f(Z) = \dfrac{4}{Z + 0.5}$, $\sigma = 4$

[TODO: Add reference and explanation for the choice of sigma and mu models]

3.3 Convolution and Sampling Effects

Having constructed the misclassification matrices for the four error models (Section 3.2), we analytically convolve the true age distribution $p_Z(z)$ to obtain the distribution of observed ages Y:

$$p_Y(y) = \sum_z M_{zy}\, p_Z(z),$$

where M is the misclassification matrix corresponding to each error model.

To assess the effects of finite sample sizes, we simulated random samples of size $n = 10, 25, 50, 100$ from the convolved distribution $p_Y(y)$. This allows us to separate variability due to sampling error from information loss caused by the convolution.

The difference between the sampled distribution and the true age distribution was quantified using Total Variation distance.

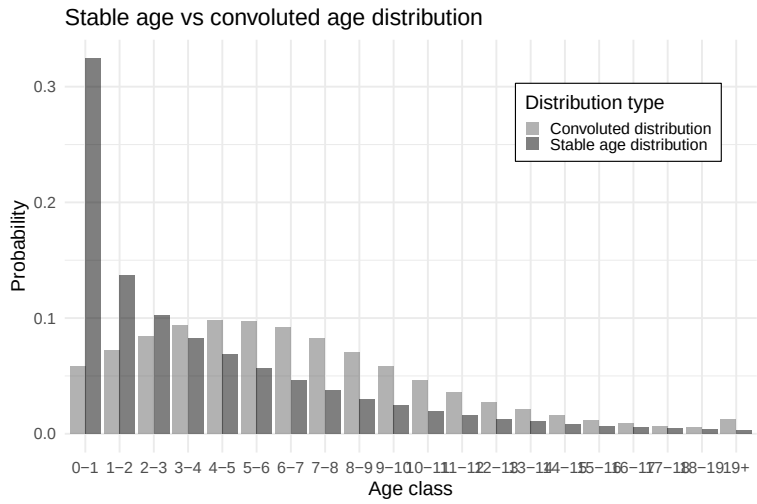


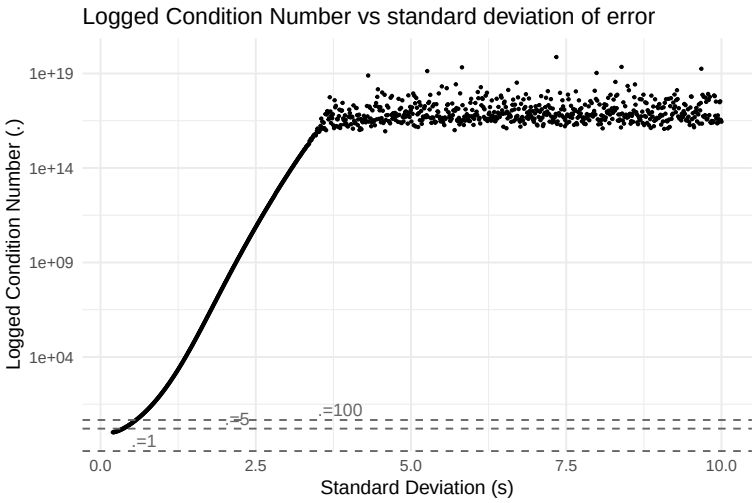
Figure 2: Stable age distribution vs convoluted age distribution using Model D

[TODO: Describe how $M^t p$ is the convoluted distribution, refer to section in Theory or write here?]

3.4 Deconvolution Methods

Two deconvolution methods were used and compared: a simple matrix inversion approach, and the EM algorithm described in Section 2.8. For each error model introduced in Section 3.2, the deconvolution was performed using both methods. The resulting estimates were evaluated using the KL-divergence, as described in Section 2.10.

Matrix inversion The matrix inversion method directly inverts the missclassification matrix that can then be used to calculate the recovered distribution, for negative estimates are set to 0 and the distribution is then normalized. While straightforward, this approach can become unstable and or ill-conditioned as σ for the error distribution increases as shown in 3.4.



EM-algorithm deconvolution We applied the EM algorithm (see Appendix **??**) using two different initializations: a uniform distribution, representing a neutral initial belief about the age distribution, and the true age distribution.

The initialization based on the true age distribution was included to assess how a well-informed prior affects the resulting estimate compared to a neutral one. In particular, this allows us to evaluate the sensitivity of the EM algorithm to initialization and to investigate for which sample sizes initialization no longer constitutes a confounding factor in an ideal setting.

The algorithm was iterated until convergence, defined as

$$\ell_c(\theta^{(t+1)}) - \ell_c(\theta^{(t)}) \leq 10^{-6},$$

or until a maximum of 100 iterations was reached.

4 Results

4.1 Effect of convolution and sampling

4.2 Effect of convolution and sampling

Figure 3 shows how sample size affects the total variation distance (TV distance) across all models. For small sample sizes, sampling noise significantly affects the TV distance. As the sample size increases, the TV distance gradually stabilises across all models. Around a sample size of approximately 100, the TV distance appears to have largely stabilised, suggesting that sampling noise plays a diminishing role relative to the structural effects induced by the convolution.

For the unbiased Models A and B, most of the variation in TV distance is driven by sampling noise, whereas for the biased Models C and D, the

TV distance is primarily driven by convolutional structure. Additionally, Model C, which is characterised by lower variance ($\sigma = 1$) compared to Model D ($\sigma = 4$), exhibits systematically higher total variation distance. This is contrary to the standard expectation that higher variance leads to more dispersed convoluted distributions and therefore larger deviations from the true distribution.

The median and non-parametric 95% upper quantile also exhibit a consistent decreasing pattern with increasing sample size, in line with the results observed for the mean (see Table 2).

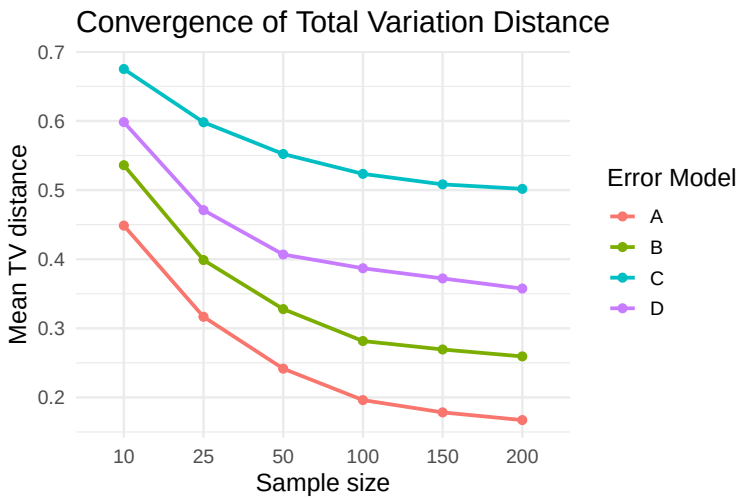


Figure 3: Mean TV distance for convoluted distributions based on Model A–D, with 100 replicas per sample size. Total variation distance is computed relative to the true distribution.

4.3 Deconvolution

Table 2 serves as a baseline for evaluating the deconvolution method.

5 Analysis / Discussion

- Discuss how errors depend on sample size and distribution parameters.
- Compare performance of EM and matrix inversion.
- Discuss practical implications for deconvolution of age structures.

Table 1: Parameters used to construct the Leslie matrix.

Age	Survival	p_coy	Litter_size	F_i
1	0.50	0.00	0.00	0.00
2	0.89	0.00	0.00	0.00
3	0.95	0.00	0.00	0.00
4	0.99	0.00	0.00	0.00
5	0.98	0.15	2.12	0.31
6	0.96	0.35	2.25	0.76
7	0.96	0.55	2.40	1.27
8	0.96	0.55	2.40	1.27
9	0.96	0.55	2.40	1.27
10	0.96	0.55	2.40	1.27
11	0.96	0.55	2.40	1.27
12	0.96	0.70	2.60	1.75
13	0.96	0.70	2.60	1.75
14	0.96	0.70	2.60	1.75
15	0.96	0.70	2.60	1.75
16	0.96	0.70	2.60	1.75
17	0.96	0.58	2.75	1.53
18	0.96	0.58	2.75	1.53
19	0.96	0.58	2.75	1.53
20	0.96	0.58	2.75	1.53

6 Appendix

6.1 Simulation Parameters

6.2 Results

model	sample_size	mean	median	upper_95
A	10	0.4486501	0.4493776	0.6160912
A	25	0.3166666	0.3115250	0.4323469
A	50	0.2416058	0.2399339	0.3251685
A	100	0.1961349	0.1898658	0.2623076
A	150	0.1783444	0.1776095	0.2312707
A	200	0.1672144	0.1671323	0.2187398
B	10	0.5360331	0.5502567	0.7117623
B	25	0.3987681	0.3871231	0.5184687
B	50	0.3277926	0.3276895	0.3944426
B	100	0.2815617	0.2785945	0.3402947
B	150	0.2693608	0.2704080	0.3151600
B	200	0.2593272	0.2577431	0.2961280
C	10	0.6752083	0.6680243	0.7934171
C	25	0.5981833	0.6023846	0.6727669
C	50	0.5522166	0.5532006	0.6114863
C	100	0.5234897	0.5194272	0.5655286
C	150	0.5082717	0.5103567	0.5431156
C	200	0.5017662	0.5005455	0.5407207
D	10	0.5983923	0.5916680	0.7658496
D	25	0.4709660	0.4693055	0.5974214
D	50	0.4067738	0.4010672	0.4894854
D	100	0.3868904	0.3827788	0.4461935
D	150	0.3721618	0.3727589	0.4154938
D	200	0.3575011	0.3567407	0.4031047

Table 2: Summary of total variation distance across models and sample sizes, based on 100 replications per configuration. Total variation distance is computed relative to the true distribution.